

Assignment 3: Gradient Descent and Newton's Method

CSED: Operations Research

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1 Objective

The objective of this assignment is to implement and compare first-order (Gradient Descent) and second-order (Newton's Method) optimization techniques. These algorithms are applied to solve a multi-variable linear regression problem predicting housing prices (SalePrice) based on above-ground living area (GrLivArea) using the Ames Housing Dataset.

2 Methodology

2.1 Model and Loss Function

The linear regression model is defined as $f(x) = a_0 + a_1x$. To evaluate the model, the Convex Sum of Squared Errors (SSE) loss function is utilized:

$$L = \frac{1}{2} \sum_{i=1}^N (y_{true}^{(i)} - (a_0 + a_1x^{(i)}))^2 \quad (1)$$

To prevent numerical instability and exploding gradients, the features were normalized using Z-score standardization prior to optimization.

2.2 Optimization Algorithms

- **Gradient Descent:** A first-order optimization algorithm that updates parameters iteratively in the opposite direction of the gradient:

$$a_i(k+1) = a_i(k) - \alpha \frac{\partial L}{\partial a_i(k)} \quad (2)$$

- **Damped Newton's Method:** A second-order algorithm that utilizes the Jacobian (J) and Hessian (H) matrices to adjust steps based on curvature:

$$X_{k+1} = X_k - \alpha H^{-1} J \quad (3)$$

3 Experiments and Results

Table ?? summarizes the experimental runs tracking the convergence of both optimization algorithms from an initial state of $a_0 = 0, a_1 = 0$.

Algorithm	Learning Rate (α)	Iterations	Final SSE Loss	Status
Gradient Descent	0.0001	62	733.177	Converged
Gradient Descent	0.01	1000	NaN	Diverged
Newton's Method	1.0	1	733.177	Converged
Newton's Method (Damped)	0.5	31	733.177	Converged

Table 1: Optimization Results Summary

3.1 Convergence Comparison

Both successful algorithms converged to the exact same optimal weight configuration: $a_0 = -0.000000$ and $a_1 = 0.706780$. However, Newton's Method achieved convergence significantly faster than Gradient Descent (1 iteration vs. 62 iterations). Because the SSE loss function for linear regression forms a perfect quadratic surface, Newton's Method can exactly model the curvature using the constant Hessian matrix and locate the global minimum instantly when $\alpha = 1.0$.

3.2 Impact of Learning Rate (α) on Gradient Descent

Experimentation with the learning rate revealed distinct behaviors:

- **Optimal α (0.0001):** Achieved steady convergence to the minimum loss of 733.177 over 62 iterations.
- **Too Large α (0.01):** Caused severe numerical instability. The steps taken were too aggressive, resulting in an immediate overshoot of the minimum. The gradients exploded toward infinity, resulting in overflow runtime warnings and a complete divergence to NaN (Not a Number) values.

4 Analysis and Visualizations

4.1 Loss Function Curves (Optimal Convergence)

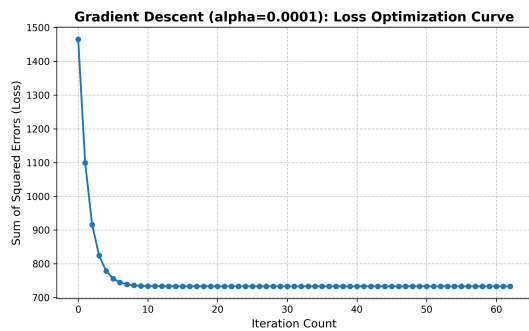


Figure 1: Gradient Descent ($\alpha = 0.0001$): Loss Curve

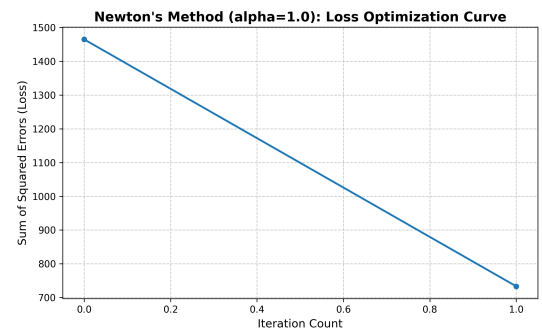


Figure 2: Newton's Method ($\alpha = 1.0$): Loss Curve

4.2 Weight Path Visualization (Optimal Convergence)

The following plots trace the trajectory of the decision variables from the initial guess to the optimum.

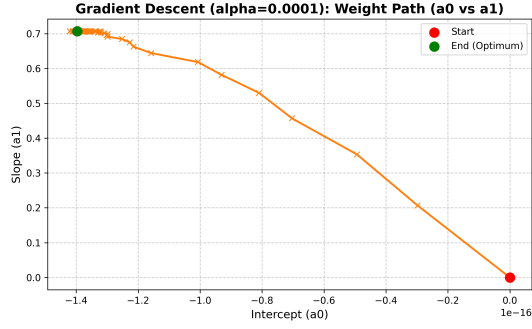


Figure 3: Gradient Descent ($\alpha = 0.0001$): Weight Path

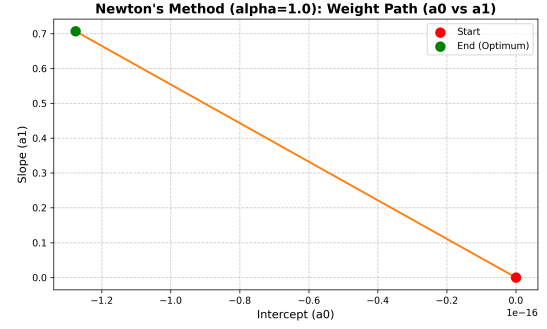


Figure 4: Newton's Method ($\alpha = 1.0$): Weight Path

4.3 Divergence Visualization ($\alpha = 0.01$)

The following plots illustrate the catastrophic failure of Gradient Descent when the learning rate is set too high ($\alpha = 0.01$). The loss rapidly explodes toward infinity, and the weight path oscillates violently out of bounds before resulting in NaN values.

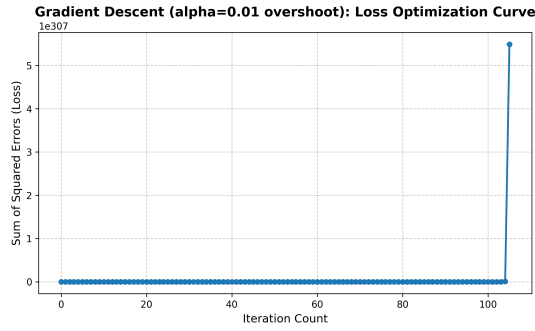


Figure 5: Gradient Descent ($\alpha = 0.01$): Diverging Loss

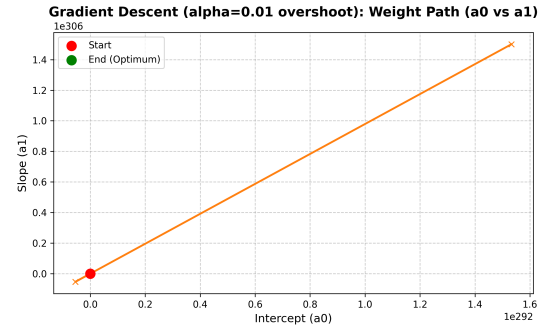


Figure 6: Gradient Descent ($\alpha = 0.01$): Diverging Weights

5 Stability Analysis

Newton's Method assumes that the local topology of the loss function can be accurately approximated by a quadratic shape. While highly efficient for the pure quadratic surface of linear regression, if applied to a complex, non-linear function with a poor initial guess, the Hessian might dictate an erratic, massive step, leading to severe divergence. A damped approach (e.g., $\alpha = 0.5$, which took 31 iterations) introduces a step size to mitigate this risk. Gradient Descent is inherently more conservative; by only following the local downward slope (first derivative) constrained by a small learning rate, it is less prone to wild jumps, making it more robust for highly irregular topologies, provided the learning rate is tuned correctly to avoid the NaN divergence observed in Experiment 2.